Hardware Setup Guide

Humanoid Robot — Raspberry Pi 4 + PCA9685

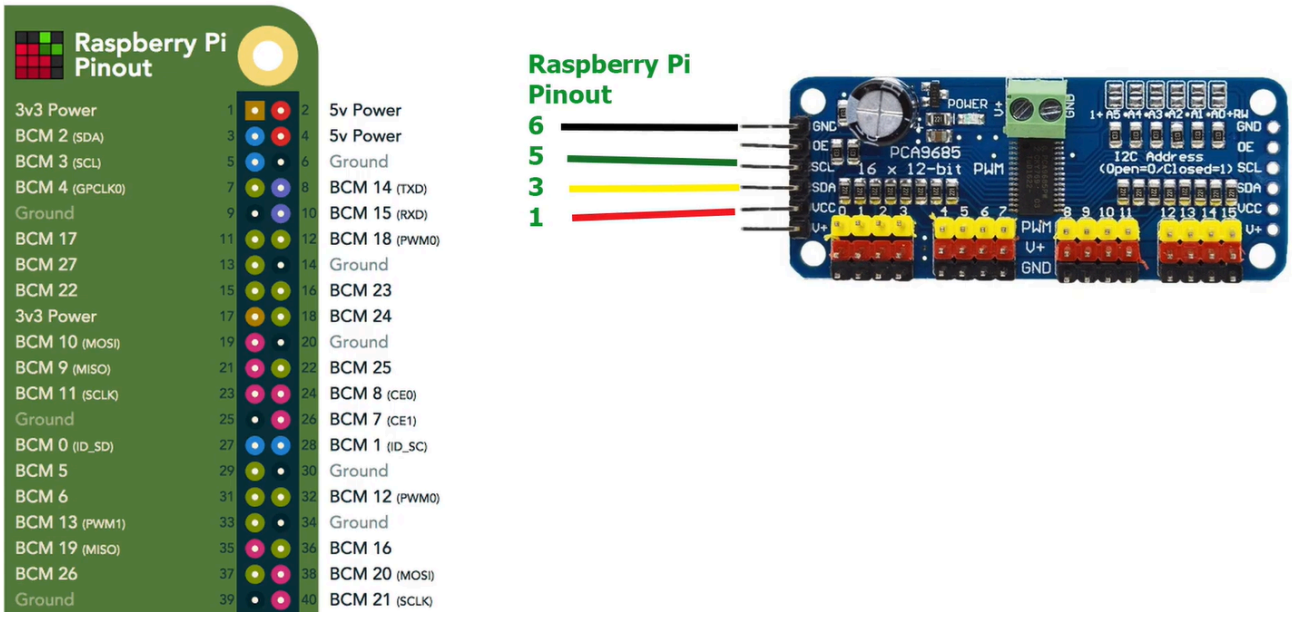
Overview

This guide covers the electrical wiring between the Raspberry Pi 4 and the PCA9685 16-channel PWM servo driver, including servo motor power supply connections.

You will need:

- Raspberry Pi 4 (any RAM variant)
- Adafruit PCA9685 16-channel 12-bit PWM servo driver
- 5V–6V regulated power supply (for servos)
- Jumper wires (female-to-female)

Section 1 — Raspberry Pi to PCA9685 (I²C)



The PCA9685 communicates over I²C. Connect four wires from the Raspberry Pi GPIO header to the PCA9685 input pins as shown below.

Wire Color	Pi Physical Pin	Pi Function	PCA9685 Pin
Red	Pin 1	3.3V Power	VCC

Black	Pin 6	Ground	GND
Yellow	Pin 3	BCM 2 (SDA)	SDA
Green	Pin 5	BCM 3 (SCL)	SCL

> **Note:** Use 3.3V (Pin 1) for VCC — not 5V. The PCA9685 logic runs at 3.3V. The V+ terminal on the board is for servo power only (see Section 2).

I²C Pin Reference (Pi GPIO Header)

Function	BCM Number	Physical Pin
SDA (Data)	BCM 2	Pin 3
SCL (Clock)	BCM 3	Pin 5
Ground	—	Pin 6 (or any GND pin)
3.3V Power	—	Pin 1 (or Pin 17)

Section 2 — Servo Motor Power Supply

Servo motors draw significant current and must be powered separately — do not power them from the Pi's 5V pins. Use a dedicated 5V–6V regulated supply connected to the PCA9685 V+ terminal block.

Wire	From	To	Notes
Power (+)	PSU positive terminal	PCA9685 V+	5V–6V regulated
Ground (−)	PSU negative terminal	PCA9685 GND	Must share ground with Pi

> **Note:** Always connect the power supply ground (GND) to the same ground rail as the Raspberry Pi. Without a common ground, signals will not reference correctly and servos will behave erratically.

Section 3 — Enable I²C on Raspberry Pi

I²C is disabled by default. Enable it before running any code.

Option A — raspi-config (recommended):

```
sudo raspi-config
```

Navigate to: Interface Options → I2C → Yes → Finish → Reboot

Option B — manual config.txt edit:

```
echo 'dtparam=i2c_arm=on' | sudo tee -a /boot/config.txt && sudo reboot
```

Verify the connection:

```
sudo i2cdetect -y 1
```

You should see address 0x40 (or 0x41–0x7F if address pins A0–A5 are bridged). If no address appears, recheck wiring.

Quick Checklist

#	Check	Status
1	SDA → Pin 3 (BCM 2)	☐
2	SCL → Pin 5 (BCM 3)	☐
3	VCC → Pin 1 (3.3V)	☐
4	GND → Pin 6	☐
5	Servo V+ connected to 5V–6V PSU	☐
6	PSU GND shares ground with Pi	☐
7	I²C enabled via raspi-config	☐
8	i2cdetect shows address 0x40	☐